Clark Zhang

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Work Authorization: Canadian Citizen — Eligible for U.S. TN Status (No H-1B sponsorship required)

EDUCATION

University of British Columbia **Graduation: April 2026**
Bachelor of Applied Science — Integrated Engineering (Mechatronics Concentration), Co-op Program
Cumulative GPA: 4.0/4.0 (92%)

SKILLS

- **Mechanical Design:** SolidWorks (CSWP), CATIA, Fusion 360, AutoCAD, SolidWorks FEA/Flow, Ansys FEA/Fluent, GD&T
- **Robotics & Controls:** Python, C++, ROS2, PLC (Siemens), UR/KUKA arms, Git, PCB design (Altium), MATLAB, RoboDK
- **Manufacturing & Prototyping:** CNC, waterjet/laser cutting, injection molding, 3D printing, composites, shop tools, drawings

WORK EXPERIENCE

Mechatronics Engineering Project Lead, Hein Lab **December 2024 – December 2025**

- Led the end-to-end development of a robotic drug-analysis platform for Canada's overdose crisis response, automating high-precision sample preparation and chemical analysis for 1,500+ real samples with 99% overall reliability
- Designed, fabricated, and tested robotic end effectors and automation modules in SolidWorks/Fusion 360, including a vial gripper, vortexer, decapper, liquid handler, and BLDC centrifuge; reduced module costs from $1,500-$5,000 to <$100
- Produced 11 shop-ready CAD assemblies and manufacturing drawings for lab-grade robotic modules, defining critical tolerances for CNC/waterjet parts, alignment interfaces, and 3D-printed components
- Built 2 custom PCBs in Altium to coordinate stepper motors, BLDC control, and sensor feedback for repeatable sample handling
- Developed a multi-tiered control architecture using Python to coordinate UR robot workflows and C++/ROS2 nodes to automate hardware modules, monitor device states, and flag sensor-reported errors

Mechatronics Engineering Intern, Hein Lab **May 2024 – December 2024**

- Commissioned 4 UR/KUKA industrial robot arms across 3 deployed automation platforms, handling mounting, I/O wiring, networking, reach validation, collision checks, and end-to-end debugging while reducing operator hands-on time by ~80%
- Transitioned rigid PLC-based workflow sequences into reusable Python control libraries for UR and KUKA arms, making robotic workflows faster to modify, test, and deploy without editing low-level control logic
- Modeled robotic workcells in SolidWorks and simulated workflows in RoboDK to verify reachability, collision clearance, and module access before deployment, reducing workflow errors by 5x

Airfoils Team Lead, UBC AeroDesign **April 2022 – April 2026**

- Led an 11-person subteam designing and manufacturing a 120-inch heavy-lift RC aircraft for SAE AeroDesign, earning 2nd place overall in 2025 out of 40+ teams
- Designed the main wing, ailerons, servo mounts, and linkage interfaces in SolidWorks and CATIA V5 using DFMA and GD&T principles, resolving clearance and manufacturability issues before final aircraft integration
- Validated aerodynamic and structural performance using SolidWorks FEA/Flow, Ansys, and wind-tunnel testing, improving lift performance by ~20% before manufacturing
- Manufactured flight hardware using 3D printing, CNC machining, waterjet/laser cutting, and carbon-fiber prepreg layups, standardizing assembly processes and reducing build errors by 70%

PROJECTS

Robotic Tool Vending Machine (Haven) **September 2025 – April 2026**

- Designed and validated a Cartesian XY gantry and rotary indexing storage mechanism for controlled checkout and return of 30 shared UBC shop tools, including pliers, cutters, and screwdrivers, reducing tool loss to zero since deployment
- Fabricated and iterated prototypes using 3D-printed parts, waterjet-cut plates, CNC-machined components, and shop tools
- Developed Raspberry Pi/Klipper motion control with magnetic encoder feedback and a custom I2C multiplexer PCB to coordinate gantry travel, storage indexing, and tool-position verification

Autonomous Diver-Support Surface Vehicle for Debris Recovery **September 2024 – April 2025**

- Developed and fabricated an autonomous diver-support surface robot with a dual-pontoon flotation chassis, adjustable propulsion mounts, and a mobile debris basket, reducing the need for repeated trips to a fixed collection point
- Built a saltwater-ready prototype with sealed plywood platforms, titanium fasteners, aluminum structural brackets, and 3D-printed propulsion/basket mounts, earning 1st place at the National CSME Design Competition